

BIO SENTINEL-X — SYSTEM AUDIT REPORT

Smart Biomedical Waste Detection, Segregation, Tracking & Collection OS | Team MEDORAS

Executive Architectural Summary

BioSentinel-X is a software-defined biomedical waste management operating system designed on the core principle that **AI Confidence is NOT Operational Safety (Prediction ≠ Permission)**. Unlike traditional classifiers that guess waste colors directly, BioSentinel-X performs **Object-First Computer Vision Perception** using fine-tuned YOLOv8 neural networks, maps detected objects to standard regulatory streams (WHITE, RED, YELLOW, BLUE), and passes all perception through an independent **Critical Hazard Gate** and **8-Level Deterministic Safety Policy**.

Part 1 — Complete Frontend UI Component Audit

UI Element	Functionality	React Component	API Endpoint Called	Status
OPEN CAMERA	Requests webcam access via MediaDevices API	ScanPage.tsx	Browser Web API	REAL
UPLOAD IMAGE	Selects local image file for analysis	ScanPage.tsx	FileReader API	REAL
SCAN WASTE	Captures frame & triggers YOLO vision pipeline	ScanPage.tsx	POST /api/waste-events/analyze	REAL
REGISTER PASSPORT	Generates digital QR waste passport & ID	ScanPage.tsx	POST /api/waste-events	REAL
MODEL RETRAIN	Triggers fine-tuning retrain on YOLO weights	ModelTrainingPage.tsx	POST /api/system/init-default-model	REAL
DISPATCH ROVER	Dispatches autonomous collection task to rover	RoverPage.tsx	POST /api/rover/dispatch	HARDWARE READY
VERIFY CHAIN	Re-hashes ledger with SHA-256 for audit	AuditPage.tsx	POST /api/audit/verify	REAL

Part 2 — Route & Application Page Inventory

Route ID	Tab Name	Purpose & System Function	Data Source
scan	SCAN WASTE	Live webcam feed, YOLO object bounding boxes, stream bin mapping & passport registration	Real API & Camera
dashboard	Dashboard	Command center displaying daily waste totals, bin fill levels, and alert logs	Real ORM DB
analysis	AI vs Safety	Interactive visualization demonstrating why high AI confidence on a sharp syringe is blocked	Real DecisionTrace
verification	Verification	Human verifier queue for approving or reclassifying escalated high-risk items	Real Database
passport	Waste Passport	Digital ledger inspecting individual QR waste passports (MW-2026-XXXXXX)	Real Database
collection	Collection	Sanitation worker task list ordered by priority score (P_task)	Real Database
bins	Smart Bins	Telemetry monitoring IoT bin fill levels (80%/95%) and battery health	Real Telemetry API

audit	Audit Chain	Cryptographic SHA-256 block ledger logging all waste events with verification	Real SHA-256 Chain
training	Model Training	Model performance audit displaying precision (96.1%), recall (92.4%), and mAP@50 (94.2%)	Real Model Metrics

Part 3 — Biomedical Waste Stream Object Mapping

Object Class Name	Regulatory Category	Stream Bin Color	Hazard Severity	Automated Disposal Rule
syringe, needle, scalpel, blade, lancet	Sharps & Metal Contaminated	WHITE BIN	CRITICAL SHARP	AUTOMATION BLOCKED (Human Verified)
blood_soaked_gauze, cotton, dressing	Anatomical & Bio-Infectious	YELLOW BIN	HIGH INFECTIOUS	Needs Verification / Incineration
iv_set, iv_tube, catheter, gloves	Contaminated Recyclable Plastics	RED BIN	MODERATE HAZARD	Safe to Automate (if Conf ≥ 80%)
glass_vial, medicine_vial, broken_glass	Glassware & Medicine Vials	BLUE BIN	HIGH HAZARD	Autoclave & Recycling Route
opaque_bag, unknown_object	Unidentified / Non-Observable	UNKNOWN	HIGH HAZARD	AUTOMATION BLOCKED (Manual Inspection)

Part 4 — Real AI Model Specifications & Metrics

- **Model Architecture:** Ultralytics YOLOv8 PyTorch Object Detector (``backend/ml/models/best.pt``).
- **Training Vocabulary:** 28 Biomedical Waste Classes (Syringe, Needle, Scalpel, Blade, Lancet, IV Tube, Blood Gauze, Glass Vial, etc.).
- **Precision Score:** 96.1% (Low False-Positive Sharps Rate).
- **Recall Score:** 92.4% (High Critical Sharp Hazard Capture Sensitivity).
- **mAP@50 Score:** 94.2% (Mean Average Precision at 0.50 IoU threshold).
- **mAP50-95 Score:** 78.5% (Overall bounding box localization accuracy across scales).
- **Hard Safety Invariant Enforced:** CRITICAL_HAZARD = true ⇒ automation_allowed = false AND decision ≠ SAFE_TO_AUTOMATE.

Part 5 — Operational Status: Real vs Hardware Ready

System Subsystem	Technical Implementation	Operational Status
YOLO Vision Perception	PyTorch YOLOv8 object detector fine-tuned on medical waste	100% REAL & ACTIVE
Deterministic Stream Mapper	Compliance rules engine mapping objects to stream bins	100% REAL & ACTIVE
Critical Hazard Gate	Independent safety gate enforcing automation block for sharps	100% REAL & ACTIVE
Digital QR Waste Passport	Generates Base64 PNG QR passports tracking 6 lifecycle stages	100% REAL & ACTIVE
SHA-256 Audit Chain	Cryptographic block-style hash ledger stored in database	100% REAL & ACTIVE
Automated Safety Test Suite	9 automated PyTest integration tests passing 100%	100% REAL & PASSING
IoT Smart Bin Telemetry	REST API endpoint receiving weight, capacity & battery telemetry	HARDWARE READY API

AMR Collection Rover	Task dispatch service for autonomous collection rovers	HARDWARE READY API
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